



Lakshya JEE 2.0 (2024)

Definite Integration

DPP-03

1. $\int_0^{\pi/2} \sin^2 x \, dx$

(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$
 (3) π (4) 0

2. $\int_0^{\pi/2} \log\left(\frac{a+b\sin x}{a+b\cos x}\right) dx =$

(1) 0 (2) $\pi/4$
 (3) $\pi/2$ (4) $\log\pi/2$

3. $\int_2^7 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{9-x}} dx$ is equal to

(1) $\frac{1}{2}$ (2) $\frac{3}{2}$
 (3) $\frac{5}{2}$ (4) 0

4. $\int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x} + e^{-\cos x}} dx$ is equal to

(1) π (2) $\frac{\pi}{2}$
 (3) 0 (4) $-\pi$

5. $\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x} dx =$

(1) $\sqrt{2}\log(\sqrt{2}+1)$ (2) $\frac{1}{\sqrt{2}}\log(\sqrt{2}+1)$
 (3) $\log(\sqrt{2}+1)$ (4) $\frac{1}{\sqrt{2}}\log(\sqrt{2}-1)$

6. $\int_0^{\pi} xf(\sin x) dx =$

(1) $\int_0^{\pi} f(\sin x) dx$ (2) $\pi \int_0^{\pi} f(\sin x) dx$
 (3) $\frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$ (4) None of these

7. Value of the integral $I = \int_0^1 x(1-x)^n dx =$

(1) $\frac{1}{n+2}$ (2) $\frac{1}{n+1} - \frac{1}{n+2}$
 (3) $\frac{1}{n+1} + \frac{1}{n+2}$ (4) $\frac{1}{n+1}$

8. If $[x]$ stands for the greatest integer function, then

$\int_4^{10} \frac{[x^2] dx}{[x^2 - 28x + 196] + [x^2]}$ is

(1) 1 (2) 2
 (3) 3 (4) 4

9. If f and g are continuous function satisfying $f(x) = f(a-x)$ and $g(x) + g(a-x) = 2$ then

$\int_0^a f(x)g(x) dx =$

(1) 0 (2) $\int_0^a f(x) dx$
 (3) $\int_a^0 f(x) dx$ (4) 1

10. $\int_{-\sqrt{3}}^{\sqrt{3}} \frac{dx}{(1+e^x)(1+x^2)}$

(1) π (2) $\frac{\pi}{3}$
 (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{2}$

11. $\int_0^{\pi} \frac{x dx}{1 + \sin x} =$

(1) $\frac{\pi}{6}$ (2) π
 (3) $\frac{\pi}{3}$ (4) None of these



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)
2. (1)
3. (3)
4. (2)
5. (2)
6. (3)

7. (2)
8. (3)
9. (2)
10. (2)
11. (2)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>